



**KAZAKH-BRITISH
TECHNICAL
UNIVERSITY**

**JSC «Kazakh British Technical University»
School of IT and Engineering**



**APPROVED BY
Dean of SITE
Azamat Amanbayev**

2025

SYLLABUS

Discipline: Software Engineering

Course code - CSCI2208

Term: Fall 2025

Instructors' full name: ALDAMURATOV Jomart, Karatayev Aron, AZIMKHANOV Bauyrzhan

Personal Information about the Instructor	Time and place of classes		Contact information	
	Lessons	Office Hours	Tel.	e-mail
Aldamuratov Jomart	L. - Monday 10:00 - 11:50 424 room	Wednesday 10:00 - 10:50		z.aldamuratov@kbtu.kz
Azimkhanov Bauyrzhan	Pr. - Monday 12:00 - 12:50 424 room			b.azimzhanov@kbtu.kz
Karatayev Aron	L. - Tuesday 10:00 - 11:50 726 room			ar.karatayev@kbtu.kz
	Pr. - Tuesday 12:00 - 12:50 726 room			
	L. - Tuesday 14:00 - 15:50 Independence Hall			
	Pr. - Tuesday 16:00 - 16:50 Independence Hall			
	L. - Wednesday 13:00 - 14:50 424 room			
	Pr. - Tuesday 15:00 - 15:50 424 room			
	L. - Thursday 9:00 - 10:50 444 room			
	Pr. - Thursday 11:00 - 11:50 444			

	room			
	L. - Thursday 14:00 - 15:50 444 room			
	Pr. - Thursday 16:00 - 16:50 444 room			

COURSE DURATION: 3 credits, 15 weeks, 45 class hours

MS Teams code (Aldamuratov J.): 5c4wnkj

<https://teams.microsoft.com/l/team/19%3AZE89CfKr8NLLZsNCy8VbEfFmvVYg0YkfnaNht82QyA1%40thread.tacv2/conversations?groupId=29790808-5232-4565-9618-dcfce8d306a4&tenantId=57081b5e-e66a-4993-8eaf-15b0b309293f>

MS Teams code (Azimkhanov B.): 7b3mdpn

<https://teams.microsoft.com/l/team/19%3AHM2GzCsMbp9kUNH7nKoN-x8y71Ykoof0fBLFhbTP2HE1%40thread.tacv2/conversations?groupId=6cf6042d-5e50-4b1d-895b-e5fde86a002f&tenantId=57081b5e-e66a-4993-8eaf-15b0b309293f>

GENERAL COURSE AIMS:

To equip students with the theoretical principles and practical skills to design, develop, and maintain robust, efficient, and ethical software systems using modern methodologies.

COURSE DESCRIPTION

This course introduces the fundamental principles of software engineering, covering the entire development lifecycle from requirements and design to implementation, testing, and maintenance of quality software systems.

COURSE OBJECTIVES

The main goal of this course is to strengthen students' knowledge and practice in the following topics:

- Introduction to Software Engineering
- Software processes & Agile software development
- Requirements engineering & Systems modelling
- Architectural design & Design, and implementation
- Software testing & Software evaluation
- Dependable systems
- Reliability engineering & Safety engineering
- Security engineering & Resilience engineering
- Software reuse & Component-based software engineering
- Distributed software engineering & Service-oriented software engineering
- Systems engineering & Systems of systems
- Real-time software engineering
- Project management & Project planning
- Quality management & Configuration management

COURSE OUTCOMES

At the end of the course, students will be able to manage & understand real-life software engineering projects.

COURSE POST REQUISITES

An overall understanding of data structures & software types is expected, even though no software coding is planned in this class

Post requisites: Artificial Intelligence, Software Testing, etc.

LITERATURE

Main - Lecture

Software Engineering, Global Edition, 10th edition, Ian Sommerville, 2016, Pearson;
<https://www.amazon.com/Software-Engineering-Global-Ian-Sommerville-ebook/dp/B01DWH1F2Y>

Software Engineering, 8th edition, Ian Sommerville, Addison wesley, 2007;
59 units are available in KBTU Library

Main - Practice

UML Distilled: A Brief Guide to the Standard Object Modelling Language, 3rd Edition, Martin Fowler, 2004, Addison-Wesley Professional;
https://www.amazon.com/UML-Distilled-Software-Engineering-Standard-dp-0582894441/dp/0582894441/ref=mt_hardcover?_encoding=UTF8&me=&qid=

Design Patterns: Elements of Reusable Object-Oriented Software, 1st edition, Erich Gamma, Richard Helm, Ralph Johnson, John Vlissides, Addison-Wesley Professional, October 31, 1994
<https://www.amazon.com/Design-Patterns-Elements-Reusable-Object-Oriented/dp/0201633612>

COURSE ASSESSMENT CRITERIA

Assessment occurs continuously throughout the course. The evaluation will be based on the levels of (maximums in %):

Type of activity	Final scores
Attendance /participation	15%
Practice	15%
SIS	22%
Midterm	8%
Quiz	0%
Final exam	40%
Total	100%

TASKS

for students' independent study (SIS)

Week	SIS	Cost (in points)
2	Assignment #01	2
4	Assignment #02	3

6	Assignment #03	3
8	Assignment #04	3
10	Assignment #05	3
12	Assignment #06	4
14	Assignment #07	4
Total		22

COURSE CALENDAR

Week	Classwork			SIS
	Topic	Lecture	Practice	TSIS
1	Course Overview Course organisation Course Importance Professional software development Software engineering ethics	1	1	
2	Software process models Process activities Coping with change Process improvement Agile methods Agile development techniques Agile project management Scaling agile methods	1	1	2
3	Functional & Non-functional requirements Requirements engineering processes Requirements elicitation Requirements specification Requirements validation Requirements change	1	1	
4	Context models Interaction models	1	1	3

	Structural models Behavioural models Model-driven architecture			
5	Architectural design decisions Architectural views Architectural patterns Application architectures	1	1	
6	Object-oriented design using UML Design patterns Implementation issues Open-source development	1	1	3
7	Development testing Test-driven development Release testing User testing	1	1	
8	Evolution processes Legacy systems Software maintenance	1	1	3
9	Dependability properties Socio-technical systems Redundancy and diversity Dependable processes Formal methods and dependability	1	1	
10	Availability and reliability Reliability requirements Fault-tolerant architectures Programming for reliability Reliability measurement Safety-critical systems Safety requirements Safety engineering processes Safety cases	1	1	3

11	Security and dependability Security and organisations Security requirements Secure systems design Security testing and assurance Cybersecurity Socio-technical resilience Resilient systems design	1	1	
12	Software reuse Component-based software engineering Distributed software engineering	1	1	4
13	Service-oriented software engineering Systems engineering	1	1	
14	Systems of systems Real-time software engineering	1	1	4
15	Project management Project planning Quality management Configuration management	1	1	
16	Final Exam			

Course assessment schedule

No	Assessment criteria	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
1	Attendance/ participation	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	15%
2	Practice	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	15%
3	SIS		2		3		3		3		3		4		4		22%
4	Midterm/ End-term							8									8%

6	Final exam																40%
	Total	2	4	2	5	2	5	10	5	2	5	2	6	2	6	2	100%

Class sessions – will be a mixture of information, discussion and practical application of skills.

Practice – will reinforce the students' knowledge by practical application of lecture materials.

In-class assessment – will prepare students for their mid-term and final assessment and identify the competence level they have achieved on a related subject matter, the aim being to diagnose potential discrepancies in students' understanding and performance to make specific adjustments to the course content and procedures and/or to assign additional assignments to certain individuals or the whole group.

Home assignments – will consolidate the concepts and materials taken during in-class activities, help students to expand the content through diverse background resources and/or practise certain skill areas; they will also develop the students' ability to work individually in exploring and examining related issues.

SIS (Student Independent Study) – comprises a group Project to be done by students on an independent basis. Students are supposed to use the knowledge and skills acquired in class to do the project. Assistance and advice will be provided by teachers during office hours.

TSIS (Teacher Supervised Student Independent Study) – student self-made project.

End-term test – a diagnostic test used to identify the students' progress, their strengths and weaknesses, intended to force students to prepare for the Final Exam. It includes computer-based test.

Final examination – 1) an attainment test designed to identify how successful the students have been in achieving objectives.

Grading policy:

Intermediate attestations (on 8th and 15th week) join topics of all lectures, practice, laboratories, SIS, TSIS and materials for reading discussed up to the time of attestation. The maximum number of points within attendance, activity, SIS, TSIS and laboratories for each attestation is 30 points.

Final exam joins and generalises all course materials, is conducted in a complex form with a quiz and problems. Final exam duration is 180 min. The maximum number of points is 40. At the end of the semester, you receive an overall total grade (summarised index of your work during the semester) according to the conventional KBTU grade scale.

ATTENTION!

- 1) If a student missed more than **30% of lessons student receives an «F (Fail)» grade.**
 - 2) If for two attestations the student receives 29.4 or fewer points, this student is not accepted to the final exam and for all course he/she receive an **«F (Fail)» grade;**
 - 3) If a student receives 9.4 or fewer points on the final exam, then regardless of how many points he/she received for the two attestations, in whole he/she receive an **«F (Fail)» grade.**
- In the case of missing or being late for the final exam without a plausible reason, regardless of how many points he (she) received for the two attestations, in whole he (she) receives an **«F (Fail)» grade.**

ACADEMIC POLICY:

1. Cheating, duplication, falsification of data, and plagiarism are not permitted under any circumstances!
2. Students must participate fully in every class. While attendance is crucial, merely being in class does not constitute "participation". Participation means reading the assigned materials, coming to class prepared to ask questions and engage in discussion.
3. Students are expected to take an active role in learning (the instructor will provide the information and guidelines to do this).
4. Students must come to class on time.
5. Students are to take responsibility for making up any work missed.

6. Make-up tests in case of absence will not normally be allowed.
7. Mobile phones must always be switched off in class.
8. Students should always show tolerance, consideration and mutual support towards other students.

Grade		Achievement percentage	Assessment criterion
«Excellent»	A	95-100%	This grade is given when the student: demonstrated a complete understanding of the course material; did not make any errors or inaccuracies; completed control and laboratory work in a timely and correct manner, and submitted reports on them; demonstrated original thinking; Submitted control quizzes on time and without any errors; completed homework assignments; engaged in research work; independently used additional scientific literature in studying the discipline; was able to independently systematise the course material.
	A -	90-94%	
«Good»	B+	85-89%	This grade is given when the student: Has mastered the course material at no less than 75%. Did not make gross errors in responses; Timely completion of control and laboratory work and submitted them without fundamental remarks; Correctly completed and timely submitted control tests and homework assignments without fundamental remarks. Utilised additional literature as indicated by the instructor; Engaged in research work, made non-fundamental errors, and fundamental errors corrected by the student themselves; Managed to systematise the course material with the help of the instructor.
	B	80-84%	
	B-	75-79%	
	C+	70-74%	
«Satisfactory»	C	65-69%	This grade is given when the student: Has mastered the course material by no less than 50%. Required assistance from the instructor when completing control and laboratory work, homework assignments.
	C-	60-64%	
	D+	55-59%	

	D	50-54%	Made inaccuracies and non-fundamental errors when submitting control tests; Did not demonstrate activity in research work, relied solely on the educational literature indicated by the instructor; Experienced more difficulty in systematising the material.
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Instructor

**ALDAMURATOV Jomart
AZIMKHANOV Bauyrzhan**

Minutes # 1 of the the School of Information Technology and Engineering meeting on August 18, 2025